



American International University- Bangladesh (AIUB)
Faculty of Engineering (EEE)

Marks
obtained:

Course Name:	Data Communication	Course Code:	COE 3201
Semester:	Spring 24-25		
Faculty:	Dr Amirul Islam	Term:	Final
Marks:	10	Assignment Name:	Assignment 01 [CO4]

CO Number	CO4	POI Number	P.f.2.C6	Knowledge Profile:	K7
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Student Name:		Student ID:		Section:	
Submission Date:		Department:	CSE		

Marking Rubrics (to be filled by Faculty):

Objectives	[2]	[1]	[0]	Secured Marks
Problem 01-5	Response shows clear systematic and theory-based understanding.	Problem is solved partially; response indicates theory knowledge is clear but fail to apply.	The solution is incorrect.	
Total				

Instructions:

- 1. Use this page as a cover page for this assignment submission. Different cover page or without cover page, assignment submission will not be acceptable.*
 - 2. The assignment should be handwritten.*
 - 3. Copied/identical submissions will be graded as 0 for all parties concerned.*
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Problem 01: We need to use synchronous TDM and combine 5 digital sources, each of 200 Kbps. Each output slot carries 1 bit from each digital source, but one extra bit is added to each frame for synchronization. **Calculate** the (a) size of an output frame in bits, (b) output frame rate, (c) duration of an output frame, (d) output data rate, and (e) efficiency of the system (ratio of useful bits to the total bits).

Problem 02: For the available bandwidth of 400 kHz, which spans from 300 to 700 kHz. **Compute** the (a) carrier frequency for half duplex mode, (b) bit rate, if modulation is done by using ASK in half-duplex mode with $d = 1$. **Sketch** the frequency spectrum for ASK in half-duplex, (c) bit rate, if modulation is done by using BFSK with $d = 0$. **Sketch** the frequency spectrum for BFSK and (d) the bandwidth for a signal transmitting at 8 Mbps for QPSK considering the value of $d = 0$.

Problem 03: We need to send 3 bits of data at a time at a rate of 9 Mbps. The carrier frequency is 20 MHz. **Compute** the number of levels (different carrier frequencies), the baud rate, and the bandwidth if QPSK is used. **Illustrate** the frequency spectrum showing the bandgap between the required carrier frequencies.

Problem 04: Six input channels, two with a data rate of 30 kbps each, one with 120 kbps, two with 60 kbps each and the last one with a data rate of 55 kbps, need to be multiplexed using synchronous TDM. A maximum of 5 kbps data can be added using the pulse stuffing method. **Illustrate** the data rate mismatch problem solution by showing suitable data rate management techniques.

Problem 05: Consider 3 ground stations located in different geographical regions, each transmitting digital data at a rate of 12 Gbps to the nearest low Earth orbit satellites via a bandwidth-limited uplink channel of 6 GHz ranging from 12 GHz to 18 GHz. (i) **Compute** the appropriate digital modulation scheme and its order ' M ' required to support the data rate of each transmitter. (ii) **Design** a suitable configuration to multiplex the four ground channels using the chosen modulation scheme and FDM. (iii) **Sketch** the constellation diagram for 8-QAM, ensuring the following:

- Properly label the axes for in-phase and quadrature components.
- Represent signal amplitudes as $\pm 3A$ and $\pm 6A$ volts.
- Clearly indicate and label all 8 constellation points.

=====THE END=====